

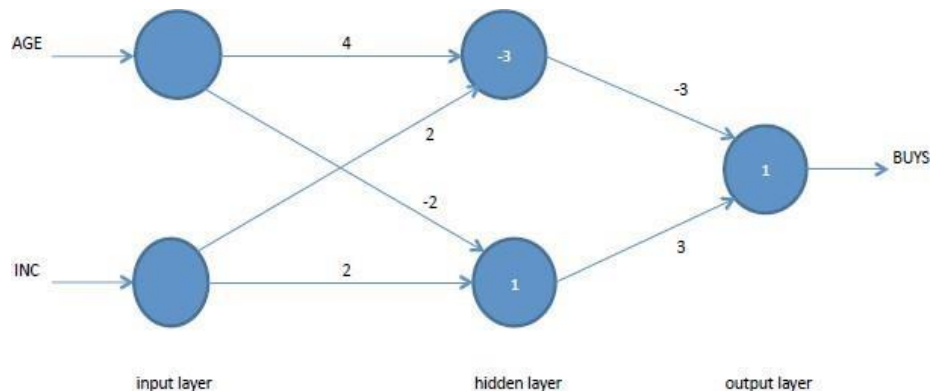
ISM 6251 and 4543: Machine Learning (Fall 2025)

Assignment 4

- Please provide succinct answers to the questions below.
- You should submit a .ipynb file and a pdf (for any non-coding questions) in Canvas.
- Please include the names of all team-members in your write up and in the name of the file. One submission per group.
- You should include all the Python functions you use.

Problem 1. Consider the problem of predicting whether a customer responds favorably to a coupon offering \$1 off the purchase price of Billy Beans. Customers can be of three types: “No” (N) customers, who will not buy Billy Beans at all, “Maybe” (M) customers, who will buy Billy Beans only with the coupon, and “Yes” (Y) customers who will buy Billy Beans whether or not a coupon is offered, but will use a coupon if given one. We will try to predict the type of customer just from knowledge of AGE and INC(ome). The continuous inputs AGE and INC have been normalized to the range 0-1 by Min-max normalization. The range of AGE is 20-80, and the range of income is 10K-110K. The output BUYS is put in the range 0-1, where 0 means NO, 0.5 means MAYBE, and 1 means YES. Operationally, any output from 0—0.2499 is NO, 0.25—0.7499 is MAYBE, and 0.75—1 is YES.

The following is the trained neural network:



Each node in the hidden and output layers is a neuron with transfer function $1/(1 + e^{-y})$, where y is the input to the node. The weights are given on each edge, and the bias is given at each node. Suppose the un-normalized inputs are AGE=70, INCOME=50K. What would be the normalized inputs (use min-max transformation), and what would be the output from the neural network?

1) Normalizing inputs

$$X_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

a. AGE = 70

$$\text{AGE}_{\text{norm}} = \frac{70-20}{80-20} = \frac{50}{60} = 0.8333$$

b. INCOME = 50,000

$$\text{INCOME}_{\text{norm}} = \frac{50,000 - 10,000}{110,000 - 10,000} = \frac{40,000}{100,000} = 0.4$$

2) Hidden layer calculations

a. Hidden neuron 1 (bias = -3)

$$y1 = (\text{AGE})(4) + (\text{INC})(2) + (-3) = (0.8333)(4) + (0.4)(2) - 3 = 1.1332$$

b. Activation function

$$\mathbf{h1} = \frac{1}{1 + e^{-y1}} = \frac{1}{1 + e^{-1.1332}} = \mathbf{0.7564}$$

c. Hidden neuron 2 (bias = -2)

$$y2 = (\text{AGE})(2) + (\text{INC})(-2) + (1) = (0.8333)(2) + (0.4)(-2) + 1 = 1.8666$$

d. Activation function

$$\mathbf{h2} = \frac{1}{1 + e^{-y2}} = \frac{1}{1 + e^{-1.8666}} = \mathbf{0.8666}$$

3) Output layer calculation (bias = +1)

a. Weights

$$\text{Yout} = (-3)(h1) + (3)(h2) + 1$$

$$\begin{aligned} \mathbf{Yout} &= (-3)(0.7564) + (3)(0.866) + 1 \\ &= (-2.2692) + (2.598) + 1 = \mathbf{1.3288} \end{aligned}$$

b. Activation function

$$\mathbf{o} = \frac{1}{1 + e^{-1.3288}} = \mathbf{0.790}$$

4) Interpretation [0–0.2499 → NO || 0.25–0.7499 → MAYBE || 0.75–1 → YES]

Since $\mathbf{0.79} \geq \mathbf{0.75}$, predicted customer type = YES

This customer is likely to buy Billy Beans even without the coupon.

Problem 2. Assume x_1 , x_2 and x_3 are three Boolean input variables (i.e. $x_1, x_2, x_3 \in \{0, 1\}$). For each of the following functions construct a neural network (draw the network) that produce the associated outputs.

(a) $(x_1 \text{ AND } x_2) \text{ AND } x_3$

(b) $(x_1 \text{ AND } x_2) \text{ OR } x_3$

(c) $(x_1 \text{ OR } x_2) \text{ AND } (x_1 \text{ OR } x_3)$

You need to determine the number of units, weights and activation functions to construct a neural network. I do not expect you to optimally compute the weights. Use any weights that you think it will give you correct results.

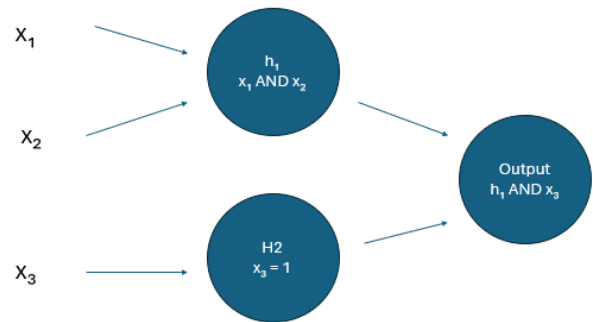
Function	Hidden Layer	Output Operation	Meaning
$(x_1 \text{ AND } x_2) \text{ AND } x_3$	1 hidden neuron (AND)	AND with x_3	All three must be 1
$(x_1 \text{ AND } x_2) \text{ OR } x_3$	1 hidden neuron (AND)	OR with x_3	Either x_3 is 1 or both x_1, x_2 are 1
$(x_1 \text{ OR } x_2) \text{ AND } (x_1 \text{ OR } x_3)$	2 hidden neurons (both OR)	AND	True if x_1 is 1, or both x_2 and x_3 are 1

(a) $(x_1 \text{ AND } x_2) \text{ AND } x_3$

- Inputs: x_1, x_2, x_3
- Weights: $w_1 = w_2 = w_3 = 1$

Gates:

- Hidden $h_1 = x_1 \text{ AND } x_2$: weights $[1,1]$, $\theta = 1.5$
- Pass x_3 directly as $h_2 = x_3$.
- Output $y = h_1 \text{ AND } x_3$: weights $[1,1]$, $\theta = 1.5$
- Fires only when all three inputs = 1.

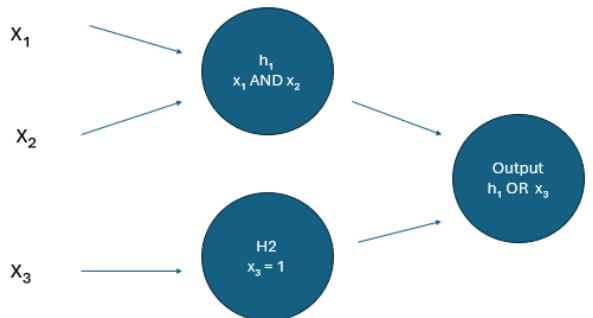


(b) $(x_1 \text{ AND } x_2) \text{ OR } x_3$

- Inputs: x_1, x_2, x_3
- Weights: $w_1 = w_2 = w_3 = 1$

Gates:

- Hidden $h_1 = x_1 \text{ AND } x_2$: weights $[1,1]$, $\theta = 1.5$
- Pass x_3 directly as $h_2 = x_3$.
- Output $y = h_1 \text{ OR } h_2$: weights $[1,1]$, $\theta = 0.5$
- Fires if either $x_3 = 1$ or both $x_1, x_2 = 1$.

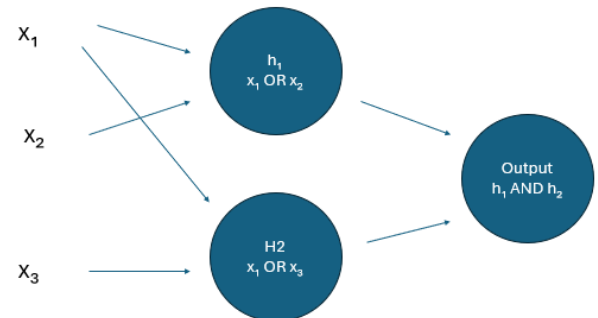


(c) $(x_1 \text{ OR } x_2) \text{ AND } (x_1 \text{ OR } x_3)$

- Inputs: x_1, x_2, x_3
- Weights: $w_1 = w_2 = w_3 = 1$

Gates:

- Hidden $h_1 = x_1 \text{ OR } x_2$: weights $[1,1]$, $\theta = 0.5$
- Hidden $h_2 = x_1 \text{ OR } x_3$: weights $[1,1]$, $\theta = 0.5$



- Output $y = h_1$ AND h_2 : weights $[1,1]$, $\theta = 1.5$
- Fires if $x_1 = 1$ or (x_2 and x_3 both = 1).

Problem 3. Assume we have data on 1000 pieces of fruit. The fruit being a Banana, Orange or some Other fruit and imagine we know 3 features of each fruit, whether its long or not, sweet or not and yellow or not, as displayed in the table below:

Fruit	Long	Sweet	Yellow	Total
Banana	400	350	450	500
Orange	0	150	300	300
Other	100	150	50	200

From this table we know

- 50% of the fruits are bananas, 30% are orange, and 20% are other fruits.
- From 500 bananas 400 are long, 350 are sweet, and 450 are yellow.
- Out of 300 oranges 0 are long, 150 are sweet and all of them are yellow.
- From the remaining 200 fruits, 100 are long, 150 are sweet and 50 are yellow.

Use the Naive Bayes model to classify the type of the fruit that is long, sweet and green.

Given Information:

- Banana = $500/1000 = 0.5$
- Orange = $300/1000 = 0.3$
- Other = $200/1000 = 0.2$

Feature Probabilities:

Banana:

- Long Probability = $400/500 = 0.8$
- Sweet Probability = $350/500 = 0.7$
- Yellow Probability = $450/500 = 0.9$
- Green Probability (not yellow) = $50/500 = 0.1$

Orange:

- Long Probability = $0/300 = 0$
- Sweet Probability = $150/300 = 0.5$
- Yellow Probability = $300/300 = 1.0$
- Green Probability (not yellow) = $0/300 = 0$

Other:

- Long Probability = $100/200 = 0.5$
- Sweet Probability = $150/200 = 0.75$
- Yellow Probability = $50/200 = 0.25$
- Green Probability (not yellow) = $150/200 = 0.75$

Naive Bayes Calculation:

Long $\times$ Sweet $\times$ Green $\times$ Fruit

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- Banana = $0.8 \times 0.7 \times 0.1 \times 0.5 = \mathbf{0.028}$
- Orange = $0 \times 0.5 \times 0 \times 0.3 = \mathbf{0}$
- Other = $0.5 \times 0.75 \times 0.75 \times 0.2 = \mathbf{0.05625}$

Normalization:

$$\text{Total} = 0.028 + 0 + 0.05625 = 0.08425$$

- Banana = $0.028/0.08425 = 0.332$
- Orange = $0/0.08425 = 0$
- Other = $0.05625/0.08425 = 0.668$

There is a 67% chance that the type of fruit that is long, sweet, and green is the “other” fruit, which is the highest of the three fruits

Problem 4. Download the cancer.csv data set (with 285 records). This data set is from the University Medical Centre, Institute of Oncology, Ljubljana, Yugoslavia and has information on the recurrence or non-recurrence of breast cancer amongst 285 patients. This data set is from a study to see if there is a recurrence of cancer based on various attributes. It includes 201 instances of one class and 84 instances of another class and 9 possible attributes (in addition to the class attribute). These are as follows (the numbers in parentheses represent the number of values that are possible for that attribute):

Class	no-recurrence-events, recurrence-events.
age (9)	10-19, 20-29, 30-39, 40-49, 50-59, 60-69, 70-79, 80-89, 90-99.
menopause (3)	lt40, ge40, premeno.
tumor-size(12)	0-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59.
inv-nodes(13)	0-2, 3-5, 6-8, 9-11, 12-14, 15-17, 18-20, 21-23, 24-26, 27-29, 30-32, 33-35, 36-39.
node-caps(2)	yes, no.
deg-malig(3)	low, medium, high breast (2) left, right.
breast-quad (5)	left-up, left-low, right-up, right-low, central.
Irradiat (2)	yes, no.

Use the Naive Bayesian method to draw up a table of probabilities for the four attributes “menopause”, “node-caps”, “deg-malig”, and “irradiat”. There’s no need to use the Python software here - just do it by hand (it might be helpful to use Excel). Based on these, what would you predict as the output classification for a woman who is premenopausal, with node-caps, a high degree of malignancy, and who has had irradiation? Show your calculation steps.

Given information:

No-recurrence = $201/285 = 0.7053$

Recurrence = $84/285 = 0.2947$

Featured Probabilities:

No-Recurrence:

- Premeno, no-recurrence = $60/201 = 0.2985$
- Yes node-caps, no-recurrence = $43/201 = 0.2139$
- 3 deg-malig, no-recurrence = $39/201 = 0.1940$
- Yes irradiate, no-recurrence = $48/201 = 0.2388$

Recurrence:

- Premeno, recurrence = $35/84 = 0.4167$
- Yes node-caps, recurrence = $28/84 = 0.3333$
- 3 deg-malig, recurrence = $30/84 = 0.3571$
- Yes irradiate, recurrence = $36/84 = 0.4286$

Naive Bayes Calculation:

Premeno × yes node-caps × 3 deg-malig × Yes irradiate

- No-Recurrence: $0.7053 \times 0.2985 \times 0.2139 \times 0.1940 \times 0.2388 = 0.002077$
- Recurrence: $0.2947 \times 0.4167 \times 0.3333 \times 0.3571 \times 0.4286 = 0.006255$

Normalization:

Total = $0.002077 + 0.006255 = 0.008332$

- No-recurrence = $0.002077 / 0.008332 = 0.2493$
- Recurrence = $0.006255 / 0.008332 = 0.7507$

For a woman who is premenopausal, with node-caps, a high degree of malignancy, and who has had irradiation we would say there is about a 75% chance of recurrence.

